

CHENSHUO-JUSTIN ZHU

☎ 778-917-6913 ✉ c23zhu@uwaterloo.ca in [LinkedIn](#) [GitHub](#) [Portfolio](#)

Education

University of Waterloo

Sep. 2023 – Present

Candidate: Bachelor of Computer Science

Waterloo, ON

- Awarded President's Scholarship of Distinction

Technical Skills

Languages: C/C++, JavaScript/TypeScript, Java, REST API, JWT, Python, C#, Bash, SQL, HTML 5, CSS 3, PHP

Frameworks: React.js, Next.js, Tailwind CSS, Node.js, Spring Boot, Spring MVC, Spring Data, JUnit

Databases & ORMs: MySQL, PostgreSQL, Drizzle ORM

Developer Tools: Git, Jenkins, Postman, VS code, IntelliJ, Cursor, Spring Tool Suite 4, Unity, Eclipse

Platforms & Environments: Apache Tomcat, Linux, Vercel

Experience

Software Developer

May 2026 – Present

Vishay Precision Group

- Building a modular **React.js** and **Tailwind CSS** UI foundation for a hot-mill crop optimization system, translating operator, maintenance, and admin workflows into role-aware browser pages.
- Improving frontend maintainability and integration readiness by standardizing shared **Storybook** components and defining backend-compatible **API/WebSocket** boundaries for future live mill-data features.

Full Stack Developer

Sep 2025 – Dec 2025

Beyz AI

- Reduced token usage by **42%** for Beyz AI's GPT-based interview assistant by iterating **prompt-engineered** compression prompts for multi-sheet user background information.
- Implemented a versioned prompt system in **Next.js** and **TypeScript** with **tRPC** and **PostgreSQL**, revalidating cached user data to prevent stale prompts and ensure consistent outputs.
- Improved **LLM cost-tracking performance** by **4×** using indexed, user-scoped **PostgreSQL** queries via **Drizzle ORM**, enabling real-time usage analytics and cost visibility.

Software Engineer

May 2025 – Aug 2025

Yoyi Internet Technology Development Co. Ltd

- Designed and developed a **Supplier Management module** within a billing system, streamlining supplier workflows and improving procurement visibility for internal teams.
- Enhanced usability and performance for over **5,000** users by building a responsive, cross-browser frontend using **React.js** and **JavaScript/TypeScript**.
- Cut API response time by **36%** by optimizing **Spring Boot REST** endpoints with **MyBatis-Plus** pagination, reducing **PostgreSQL** reads and payload size; validated via **Postman**.

Presentation Technologies Support Technician

Sep 2024 – Dec 2024

Information Systems & Technology Department, University of Waterloo

- Improved Board & Senate rooms seat assignment efficiency by **75%** by developing an attendee management web system with **PHP**, **JavaScript**, and **PostgreSQL** using an interactive **SVG** seating plan.
- Delivered high-touch AV support across **90%** of campus classrooms, resolving **350+** issues by communicating effectively with professors and providing patient step-by-step guidance.

Projects

ImageNest | Website [🔗](#), GitHub [🔗](#)

May 2025 – Aug 2025

- Built a dynamic image gallery in **Next.js** with **App Router** and **modals**, improving transitions and navigation.
- Implemented access control with **Clerk** and optimized cold-starts via caching and rate limiting (**Upstash Redis**).
- Enabled secure uploads and efficient rendering using **UploadThing** and **PostgreSQL** in a server-rendered stack.

Sorcery - Card Game

May 2024 – Aug 2024

- Collaborated in a 3-person team to develop a turn-based card game engine using **C++** and **OOP principles**.
- Utilized **UML** to plan game and implemented modular card components within **MVC pattern** for flexible extension.
- Partnered with teammates to integrate and debug match flow and player actions, ensuring smooth gameplay.

Interactive Museum Guide Website | Website [🔗](#), GitHub [🔗](#)

Sep 2022 – Jun 2023

- Constructed an interactive museum guide website using **HTML**, **CSS**, and **JavaScript** with unique features.
- Enhanced artifact viewing experience by implementing a magnifying tool within a creative world-map interface.